Lorenzo Campoli

Date of birth: 05/04/1986
Nationality: Italian
Email: campoli.lorenzo@gmail.com
GitHub: https://github.com/lkampoli
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Research Areas

Scientific computing, High-Performance Computing (HPC), Computational Fluid Dynamics (CFD), Hypersonics, Thermo-chemical Non-equilibrium flows, Magnetohydrodynamics, Numerical methods for shock flows, Shock-Fitting, Data-driven turbulence modelling, Machine learning for fluid dynamics, Space rocket propulsion and aerothermodynamics, Transformer models, Probabilistic programming, Bayesian inference, Clustering algorithms, High-dimensional optimization.

Current Employment

Senior System Modelling Engineer Sept. 2024 – present
Gilmour Space Technologies https://www.gspace.com

- Determination of aerodynamic properties and aerodynamic heating of launch vehicles from subsonic to hypersonic flow regimes using CFD and other analytical methods
- Aeroelastic and structural loads analysis for launch vehicles
- Developing and maintaining in-house engineering tools for aerodynamic and structural dynamics analysis including coupled loads analysis.

Research Fellow May 2022 – June 2024
University of Melbourne, Faculty of Engineering and Information Technology
https://energy.unimelb.edu.au

- Data-driven turbulence closure modeling
- Gene Expression Programming enhancement with clustering

Associate Professor June 2023 – present
Macquarie University, Faculty of Engineering https://www.mq.edu.au

- Teaching the course of Fluid Mechanics

Professional Development

CSIRO ON Prime 15 Program Apr. 2024 – Jun. 2024
https://research.csiro.au/onalumni/hypersocks
https://hypersocks.quarto.pub/hypersocks-team

Previous Work Experience

Assistant Professor Jun. 2020 – Jun. 2022
Saint Petersburg State University, Faculty of Mathematics and Mechanics http://gam.spbu.ru

- Research: CFD and machine learning code development for non-equilibrium flows in state-to-state and multi-temperature formulations
- Teaching activity (see Teaching section)

Assistant Researcher Nov. 2018 – Apr. 2020
Saint Petersburg State University, Faculty of Mathematics and Computer Science https://chebyshev.spbu.ru

- Numerical modelling of viscous finger phenomenon
- Enhanced recovery methods (EOR) for flows in porous media

Supervisors: Prof. S. Tikhomirov

PostDoc Researcher Nov. 2017 – Nov. 2020
Saint Petersburg State University, Faculty of Mathematics and Mechanics http://gam.spbu.ru

- Code development for high-speed non-equilibrium reacting flows with state-to-state approach

Supervisors: Prof. E. Kustova

Visiting Researcher Jun. – Jul. 2015
NASA Ames Research Center, Mountain View, CA, USA

- Implement shock-fitting technique onto the high-order finite-difference NASA code ADPDIS3D

Supervisors: Dott. H. C. Yee, Dott. D. Kotov, Prof. B. Sjogreen

Visiting Researcher Mar. – Jun. 2015
INRIA Bordeaux, France www.inria.fr/centre/bordeaux

- Coupling of shock-fitting algorithm with RD INRIA code RD-RK2 for unsteady 2D flows on moving/deforming unstructured grids (ALE formulation)

Supervisors: Dott. M. Ricchiuto, Prof. R. Paciorri

Research Assistant May – Dec. 2013
AVIO, Colleferro, Italy www.aviogroup.com

- Pressure oscillations numerical simulations of P80 Vega Solid Rocket Motor (SRM)
- Aerothermodynamic numerical simulations on Vega launcher fairing

Supervisor: Dott. F. Paglia

Technical Consultant Feb. – Apr. 2013
NHAZCA, Rome, Italy www.nhazca.it

- Algorithm development for terrestrial and satellite interferometric radar (SAR, TinSAR, DinSAR)

Stagist Jun. – Jul. 2004
INFN, Frascati, Italy www.infn.it

- Homogeneity measurements of a MultiWire Proportional Chamber (MWPC)

Supervisor: Dott. M. Anelli

Teaching Experience

Saint Petersburg State University, Faculty of Mathematics and Mechanics https://gam.spbu.ru

- Hypersonics Sep. 2020 – Sep. 2022
- Machine Learning for Fluid Mechanics Mar. 2021 – Sep. 2022
- Modern Scientific Visualization Mar. 2021 – Jun. 2021
- Scientific Paper Writing Mar. 2021 – Jun. 2021
- Concepts of Modern Natural Science Mar. 2021 – Jun. 2021

Degrees

Ph.D. in Theoretical and Applied Mechanics Nov. 2013 – Nov. 2016
Sapienza University of Rome (Dept. of Mechanics and Aeronautics)
Thesis: A shock-fitting technique for two-dimensional unsteady flows on unstructured grids
Defence: 08/02/2017
Supervisor: Prof. R. Paciorri
Advisor: Prof. C. M. Casciola

Second-Level Master in Space Transportation Systems Feb. 2012 – Dec. 2012
Sapienza University of Rome https://web.uniroma1.it/mastersts
Thesis: Solid rocket motor's internal fluid dynamics: fluid-structure interaction
Defence: 20/01/2013
Supervisor: Dott. F. Paglia
Advisor: Prof. F. Nasuti

Master in Space Engineering Sept. 2008 – Feb. 2012
Sapienza University of Rome
Thesis: Modeling and simulation of base flows in subsonic regime
Defence: 20/02/2012
Supervisor: Prof. R. Paciorri

Bachelor in Aerospace Engineering Sept. 2005 – Dec. 2008
Sapienza University of Rome
Thesis: Ablative materials: types and characterization
Defence: 20/12/2008
Supervisor: Prof. G. Rinaldi

Technical Skills

OS	Linux (OpenSuse, Mint, Debian, Ubuntu, Kali)
Programming languages	Fortran, C/C++, Matlab, Mathematica
Parallel programming	MPI, OpenMP, Coarray, CUDA, OpenACC
Profiling & debugging	gprof, perf, gdb, Valgrind, AddressSanitizer, Vtune, Advisor, Paraver, Scalasca, Extrae, LIKWID, PAPI
Version control	svn, bzr, git
Scripting	Bash, Perl, Python, Jupyter, Colab, R
ML & data mining	TensorFlow, Keras, PyTorch, Pandas, Scikit-learn, NumPy
Document processors	LaTeX, Lyx, Office, Overleaf
CFD solvers	STAR CCM+, CFD++, Comsol, Dumux, Fluent, Ansa, OpenFOAM, Nektar++, SU2, COOLFluid, FronTier++, MRST, Overture, in-house codes (see GitHub page)
Mesh generators	Triangle, Tetgen, Delaundo, Yams, Gmsh, GRUMMP, mmg3d, MeshLab, Gambit, Fluent Mesh
Visualization	Tecplot, Gnuplot, VisIt, ParaView, Matplotlib, Blender
CAD	AutoCAD, FreeCAD, SolidWorks, Catia, SpaceClaim

Language Skills

Italian (mother tongue), English proficiency (C1), upper intermediate Russian (B2), intermediate Spanish (A2), intermediate French (A2), Chinese (HSK 3).

Research Outputs

L. Campoli, A. Assonitis, M. Ciallella, R. Paciorri, A. Bonfiglioli, M. Ricchiuto. UnDiFi-2D: an Unstructured Discontinuity Fitting code for 2D grids. *Computer Physics Communications*, 2021. doi:10.1016/j.cpc.2021.108202 **IF=3.4, Q1**.

L. Campoli, Kustova, E., Maltseva, P. (2021). Assessment of machine learning methods for state-to-state approaches. *arXiv:2104.01042*.

L. Campoli, Kunova, O., Kustova, E., Melnik, M. (2020). Models validation and code profiling in state-to-state simulations of shock heated air flows. *Acta Astronautica*. doi:10.1016/j.actaastro.2020.06.008 **IF=3.4, Q1**.

Bakharev, F., Campoli, L., Enin, A., et al. (2020). Numerical Investigation of Viscous Fingering Phenomenon for Raw Field Data. *Transport in Porous Media*, 1–22. doi:10.1007/s11242-020-01400-5 **IF=2.6, Q1**.

L. Campoli, G. Oblapenko, M. Mekhonoshina, E. Kustova. Numerical investigation of hypersonic non-equilibrium flow around blunt body by COOLFluiD-Kappa coupling. *AIP Conference Proceedings*. RGD31, Glasgow, UK, 2018.

L. Campoli, G. P. Oblapenko and E. V. Kustova. KAPPA: Kinetic Approach to Physical Processes in Atmospheres library in C++. *Computer Physics Communications*, 2018. doi:10.1016/j.cpc.2018.10.016 **IF=3.4, Q1**.

L. Campoli, Quemar, P., Bonfiglioli, A., & Ricchiuto, M. (2017). Shock-fitting and predictor-corrector explicit ALE Residual Distribution. In *Shock Fitting* (pp. 113–129). Springer, Cham. (Book chapter).

A. Bonfiglioli, R. Paciorri, L. Campoli. Unsteady shock-fitting for unstructured grids. *Int. J. Numerical Methods in Fluids*, 2015. doi:10.1002/fld.4183 **IF=2.6, Q1**.

Conferences

L. Campoli, R. D. Sandberg, Y. Fang, X. Xu. Enhancement of gene expression programming for RANS turbulence modeling with unsupervised clustering. *23rd Australasian Fluid Mechanics Conference*, Sydney, Dec. 2022.

L. Campoli, Oblapenko, G. P., & Kustova, E. V. (2019). Overview and perspectives of KAPPA library. *AIP Conf. Proc.* (Vol. 2132, p. 150005). doi:10.1063/1.5119645

L. Campoli, G. Oblapenko, M. Mekhonoshina, E. Kustova. Hypersonic non-equilibrium flow around blunt body by COOLFluiD-Kappa coupling. RGD31, Glasgow, 2018.

O. V. Kunova, L. Campoli. On the implementation of the software library Kappa and its interface with COOLFluiD. *Eighth Polyakhovs Reading*, Saint Petersburg, 2018.

A. Bonfiglioli, R. Paciorri, L. Campoli, et al. Development of an unsteady shock-fitting technique for unstructured grids. ISSW30, Tel-Aviv, Jul. 2015.

A. Bonfiglioli, R. Paciorri, L. Campoli. Unstructured shock-fitting calculations of transonic turbomachinery flows. ETC11, Madrid, Mar. 2015.

A. Bonfiglioli, R. Paciorri, L. Campoli (2014). An unsteady shock-fitting technique for unstructured grids. *ECFD VI*, Barcelona.

M. Onofri, R. Paciorri, L. Campoli, A. Bonfiglioli. An unsteady shock-fitting technique for unstructured grids. ISIS21, Riga, Aug. 2014.

R. Paciorri, L. Campoli. Numerical Simulations of Flows Past the IXV Capsule. *5th Int. ARA Days*, Arcachon, May 2015.

Research Funding & Grants

Dog_2019 *13 Mar – 16 May 2019*

Development of tools for determining optimal EOR: connection of EOR and mechanisms of influence, methods of monitoring

Funded by: **Gazprom Neft**

RSF_RG_2019 – Stage 1 *6 May – 31 Dec 2019*

Modeling of non-equilibrium carbon dioxide flows in modern problems of space aerodynamics and ecology of the Earth

Funded by: **Russian Science Foundation**

Dog_2019 *31 May – 30 Sep 2019*

Development of methods for increasing the predictive ability of three-dimensional digital geological models

Funded by: **Gazprom Neft**

Dog_2019 *22 Oct – 31 Dec 2019*

Development of tools for determining the optimal EOR in terms of SAW-polymer and alkaline-polymer flooding, estimates of Kohv in a 5-point system; evaluation of the optimal polymer fringe size

Funded by: **Gazprom Neft**

Dog_2019 *16 Dec 2019 – 19 Mar 2020*

Development of methods for increasing the predictive ability of three-dimensional digital geological models

Funded by: **Gazprom Neft**

RSF_RG_2019 – Stage 2 *1 Jan – 31 Dec 2020*

Modeling of non-equilibrium flows of carbon dioxide in modern tasks of space aerodynamics and ecology of the Earth

Funded by: **Russian Science Foundation**

Dog_2019 – Donation No. 117/19-BP *1 Jan – 31 Dec 2020*

Donation agreement

Funded by: **Foundation for Support of Social Initiatives "Rodnye Goroda"**

RSF_RG_2019 – Stage 3 *1 Jan – 31 Dec 2021*

Modeling of non-equilibrium flows of carbon dioxide in modern tasks of space aerodynamics and ecology of the Earth

Funded by: **Russian Science Foundation**

M1_2021 – Stage 1 *24 Aug – 31 Dec 2021*

Machine learning in problems of non-equilibrium aeromechanics

Funded by: **Saint Petersburg State University**

Awards & Honours

Certificate of participation in several PRACE Training Centre's courses.
https://training.prace-ri.eu

Partnership of the European Group of Aeronautics and Space UniversitieS (PEGASUS), 2012.
https://www.pegasus-europe.org